

Question 1 : What is Power BI and why is it used in businesses?

1.

Ans - Definition: A tool that connects to multiple data sources, cleans and transforms data, and presents it through interactive dashboards and reports.

Why Businesses Use Power BI

- **Data Visualization:** Turns complex datasets into clear charts, graphs, and dashboards.
- **Real-Time Insights:** Automated refreshes keep reports up to date.
- **Collaboration:** Teams share dashboards across departments, ensuring everyone works with the same data.
- **Efficiency:** Reduces manual reporting, saving time and minimizing errors.
- **Scalability:** Handles large datasets and supports enterprise-wide usage.

Question 2 : Name and explain the three main components of Power BI .

Ans- Components:

1. Power BI Desktop

- **What it is:** A free Windows application used to design and build reports.
- **Purpose:** Allows users to connect to data sources, clean and transform data, create data models, and design interactive visualizations.

- Use Case: Analysts use Power BI Desktop to prepare dashboards before publishing them to the cloud service.

2. Power BI Service (Cloud)

- What it is: An online SaaS (Software as a Service) platform hosted by Microsoft.
- Purpose: Enables sharing, collaboration, and distribution of reports and dashboards across teams and organizations.
- Use Case: Managers and employees access published dashboards in real time, ensuring everyone works with the same updated data.

3. Power BI Mobile

- What it is: Mobile apps available for iOS and Android devices.
- Purpose: Provides on-the-go access to dashboards and reports.
- Use Case: Executives and field staff can check KPIs and performance metrics anytime, anywhere.

Question 3 : Explain the Power BI workflow.

Ans - The Power BI workflow moves from connecting data → transforming → modeling → visualizing → sharing → decision-making. This structured process ensures businesses get accurate, interactive insights that drive smarter decisions.

Question 4 : List any four data cleaning tasks that can be performed in Power Query

Ans - Power Query helps clean and shape data before analysis. Tasks like removing duplicates, handling missing values, changing data types, and splitting/merging columns ensure the dataset is accurate and ready for visualization in Power BI.

Question 5 : Write step by step instructions to load the above dataset.

To load the Real_Estate_Data_100_Rows.csv dataset into Power BI

Step 1- Open Power BI Desktop

Step 2- Get Data

Home → Get Data → Text/CSV

Step 3- Locate the File

Step 4- Preview and Load

Check the preview window to confirm the data looks correct.

Step 5- Transform Data (Optional)

Data Cleaning

Use Power Query to clean and shape the dataset before loading.

Step 6- Finalize and Use

Once loaded, the dataset is ready for visualization.

Question 6: Define Data View, Report View, and Model View. Explain the purpose of each view.

Ans - Explanation of the three main views in Power BI and their purposes:

1. Data View

- **Definition:** Displays the underlying data tables after they've been loaded and transformed.
- **Purpose:**
 - Allows you to inspect rows and columns directly.
 - Useful for verifying data quality, checking transformations, and creating calculated columns using DAX.

Question 7 : Discuss the different data sources that Power BI supports.

Ans -

Category	Examples	Purpose
Flat Files	Excel, CSV, JSON, PDF	Local structured/semi-structured data
Databases	SQL Server, Oracle, MySQL	Relational data, real-time queries
Azure Cloud	Azure SQL, Synapse, Blob Storage	Cloud-native integration
Online Services	Salesforce, Google Analytics, Dynamics 365	SaaS and CRM data
Other Sources	Hadoop, Spark, ODBC	Big data and enterprise systems

Question 8 : Split Owner Name to create two new columns as First Name and Last Name

Ans-

Step 1- Load the Dataset

Step 2- Open Power Query Editor

Step 3- Select the Column

Step 4- Apply Split Column

Step 5 - Rename the New Columns

Step 6- Close & Apply